Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/08/2004
PI Name:	Thorne, Robert E. Ph.D.		
Project Title:	Macromolecular Crystal Growth: Techniques Development, Fundamental Studies, and Applications to Complexes Affecting Cell Growth and Division		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Ithaca	**State:**	NY
Zip Code:	14853-2501	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/30/2001	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ellenberger, T. (Harvard Medical School) Duffy, T. (Geosciences Princeton University)		
Grant/Contract No.:	NAG8-1831		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Growth of high quality macromolecular crystals is a major factor limiting the rate at which new macromolecular structures are determined. We are pursuing an interdisciplinary experimental program to advance fundamental understanding of macromolecular crystals and to solve specific structures important in cell growth and division. Our program involves two major components: (1) Development and application of novel or improved experimental probes of macromolecular solutions and crystals, including X-ray imaging techniques, SAXS, and Brillouin scattering; and (2) Application of these and other techniques to help obtain high-quality crystals of DNA helicase-primase complexes and CDC42 and its regulators. The objectives of this work are the development of novel or improved experimental probes of macromolecular solutions and crystals, and specific attacks on crystal-related problems of molecular complexes involved in cell growth. New/improved techniques for characterizing macromolecular solutions and crystals should yield		

Task Description:	new insights into intermolecular interactions, crystal disorder, and molecular structure. Applying these and our current techniques (so far restricted to simple model proteins) to macromolecular complexes of significant biological interest will help to identify and refine a subset of these techniques most useful to the structural biology community. The graduate students involved in this program will receive an interdisciplinary training well suited to addressing future challenges in this expanding area. This program will assist NASA's microgravity effort by developing new methodologies for evaluating the success of flight experiments, by testing specific hypotheses about defect formation relevant to the microgravity environment, and by helping to disseminate the results from its microgravity crystal growth program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work should significantly impact the general practice of macromolecular crystallography, and thus the fundamental understanding of biological processes and the development of new medicines.
Task Progress:	In the past year we have continued the development of our new technology for han-dling and mounting crystals for cryocrystallography, which was begun under this grant and since Fall 2002 has been supported by the NIH. Our method is based on replacing the nylon loops currently used with a thin, microfabricated polyimide structure. We have provided approximately 2000 of the mounts to more than 150 individuals and groups worldwide. The response has been enthusiastic. One group was able to obtain a com-plete data set from a 5 micrometer crystal at the Swiss Light Source using our mounts. A poster describing the mounts was awarded the Oxford Cryosystems Prize at the 2003 American Crystallographic Association annual meeting. Based on feedback we have re-ceived, we made several improvements. Cornell has applied for a patent, and the PI has established a company - MiTeGen, LLC - to manufacture and distribute the mounts, with an initial shipping target date of July 1, 2004. We are continuing to develop a range of related technologies. We have also designed and built an inexpensive temperature con-trolled incubator for crystal growth that provides better than 0.1 °C temperature stability and very low vibration levels. Currently there are no commercially available products for under $2000, and our incubator provides better performance for $300. We are gener-ating a web page explaining the construction so that they can be reproduced by the mac-romolecular crystallography community. We have continued our exploration of surface chemical treatments of crystallization ap-paratus to address two problems. The first is that crystals tend to stick to the surfaces onto which they grow, and this makes them difficult to remove for diffraction studies without damaging them. The second is that protein crystallization drops tend to have irregular shapes because of contact line pinning during drop dispensation and because of drop movements when, e.g., drops are flipped for hanging drop crystallization. Since the rate of equilibration of the drop with the well is, for a given drop volume, determined by the drop shape, this is almost certainly an important source of variability in crystallization experiments. We have developed surface chemical treatments that provide strong contact line pinning at the boundary of the drop and thus ensure drop shape reproducibility. We have designed and built an apparatus to allow time-resolved studies of both static and ac-celerated (vibrating) drops to further quantify the effects of contact line pinning on drop shape. Vibrating drops simulate the effect of accelerations produced during drop dispen-sation and during manipulations such as drop inversion for hanging drops. Based on these results we have filed a patent disclosure on a fundamentally new kind of crystalliza-tion plate compatible with both ground and microgravity based experiments. Our design is compatible with standard equipment for high-throughput screening. At the same time it gives the kind of precise control over crystallization possible with Fluidigm's microflu-idic technology in a much more convenient way and at much lower cost. We will con-tinue development of this technology in the coming year. The insights gained in our work on flash cooling of protein crystals have spawned a new research project on freezing of biological cells. We are expanding our collaboration with Prof. Alex Travis of Cornell's Department of Veterinary Medicine to develop improved protocols for freezing sperm of endangered species, an important tool in maintaining bio-diversity. Our eventual goal is to extend methods developed to cryopreservation of other cells which will be important, for example, in long-term human space flight. In the past year we have designed improved apparatus for freezing cells, and will file a preliminary patent application on this technology shortly. A Master of Engineering student has been hired to construct and test this apparatus. We have also continued our work on AFM studies of structure and disorder of T7 helicase-primase crystals. The Harvard group continues to perform the production of the complex, and the Cornell group now does the crystallization. Preliminary results show that the crystals grow without dislocations or vacancies, and attempts to improve the im-age resolution to look for multiple conformers are in progress.
Bibliography Type:	Description: (Last Updated: 03/14/2017)
Articles in Other Journals or Periodicals	Stum Z, Kmetko J, O'Neill K, Gillilan RE, Thorne RE. "A new crystal mounting method for macromolecular crystallography" Synchrotron Radiation News. 2004;17(2):31-2. http://dx.doi.org/10.1080/08940880408603083 , Sep-2004
Articles in Peer-reviewed Journals	Speziale, S.; Jiang, F.; Caylor, C.L.; Kriminski, S., Zha C.S.; Thorne, R.E.; Duffy, T.S. "Sound velocity and elasticity of tetragonal lysozyme crystals by Brillouin spectroscopy" Biophys J. 85 (2003) 3202-3213 , Oct-2003
Articles in Peer-reviewed Journals	Thorne, R. E.; Stum, Z.; Kmetko, J.; O""Neill, K.;Gillilan, R. "Microfabricated mounts for high-throughput macromolecular cryocrystallography" J. Appl. Cryst. 36 (2003) 1455-1460 , Dec-2003
Awards	Stum, Z.; Kmetko, J.; O""Neill, K.; Gillilan, R. L.; Thorne, R. E. "Oxford Cryosystems Low Temperature Prize, 2003 ACA Annual Meeting" Jul-2003
Dissertations and Theses	Kriminski, S. "Disorder in flash-cooled protein crystals" Ph.D. dissertation May-2004
Patents	Not assigned Application filed March 2004 Jan-2004 Thorne, R. E.; Stum, Z.; O'Neill, K.; Kmetko, J. "Sample Mounts for Microcrystal Crystallography"

Presentation	Thorne, R.E.; Malkin, A. "Growing larger protein Crystals: Some insights from fundamental studies" NIGMS Protein Production Workshop, Bethesda, MD, March 29-31, 2004. Mar-2004